

BEAUFORT COUNTY AP, SC, USA

WMO: 720612

Lat: 32.412N

Lon: 80.634W

Elev: 125

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00203

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.4	32.5	12.3	10.3	36.2	18.1	13.7	40.0	23.2	46.6	20.8	49.2	8.0	10	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	93.4	77.1	91.0	76.9	90.0	76.8	80.5	87.4	79.6	86.4	78.8	85.4	7.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	151.2	83.1	77.4	143.3	82.3	77.2	142.0	82.2	44.0	88.7	43.3	85.8	42.5	85.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	17.9	15.8	27.1	97.3	3.6	1.5	24.5	98.4	22.4	99.3	20.4	100.1	17.8	101.2
			23.1	81.9	3.1	1.1	20.9	82.7	19.1	83.4	17.3	84.0	15.1	84.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.4	49.8	53.9	58.3	66.9	74.3	80.7	82.6	81.7	77.9	69.3	58.6	54.1	(6)
	DBStd	13.19	8.97	8.51	7.96	6.03	4.85	3.73	2.96	2.77	3.81	6.48	7.46	8.23	(7)
	HDD50	255	115	53	22	0	0	0	0	0	0	0	13	52	(8)
	HDD65	1679	474	319	232	51	4	0	0	0	0	34	219	346	(9)
	CDD50	6609	110	161	279	506	752	922	1009	982	838	599	270	181	(10)
	CDD65	2556	3	7	24	106	291	472	544	517	388	168	26	10	(11)
	CDH74	22598	6	29	117	510	1911	4710	5937	5280	3080	947	62	9	(12)
	CDH80	7571	0	0	17	60	464	1761	2343	1882	887	156	1	0	(13)
(14) Wind	WSAvg	6.4	7.0	7.0	7.4	7.2	6.4	5.6	5.5	5.2	5.8	6.2	6.6	6.6	(14)
(15) Precipitation	PrecAvg	47.6	3.3	3.2	3.5	2.9	2.8	5.1	5.5	6.4	4.8	4.6	2.5	3.0	(15)
	PrecMax	67.7	8.7	11.1	8.6	7.2	7.2	12.6	10.1	16.3	9.3	20.9	5.6	10.5	(16)
	PrecMin	31.9	0.6	0.3	0.3	0.6	0.5	1.5	1.2	2.0	0.7	0.2	0.0	1.0	(17)
	PrecStd	8.5	2.2	2.2	1.9	1.9	1.9	2.5	2.4	3.0	2.1	4.0	1.8	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	77.3	82.0	86.1	90.8	96.8	96.6	95.2	90.7	86.4	79.1	75.0	(19)
		MCWB	65.1	65.0	63.7	67.5	70.9	77.0	77.7	78.5	75.7	75.3	69.4	68.7	(20)
	2%	DB	70.1	73.1	77.1	81.3	88.1	93.2	93.3	91.2	88.4	83.7	75.4	72.3	(21)
		MCWB	63.2	63.4	64.0	68.3	71.0	76.3	77.9	77.8	75.7	74.7	67.8	67.1	(22)
	5%	DB	65.9	70.3	72.9	79.2	84.5	90.6	90.8	90.1	87.6	81.5	72.7	70.0	(23)
		MCWB	59.8	62.1	62.1	67.0	71.6	76.2	77.7	77.8	75.3	72.0	66.3	66.5	(24)
	10%	DB	63.3	66.4	70.3	77.1	81.8	88.3	89.8	88.2	85.8	79.3	70.4	67.7	(25)
		MCWB	58.6	60.5	61.3	66.2	71.0	75.8	77.6	77.2	74.9	70.6	65.3	64.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.1	68.9	69.5	73.8	76.4	81.3	81.4	81.3	80.1	78.3	73.7	72.2	(27)
		MCDB	70.8	73.3	74.7	79.0	83.6	89.3	89.2	87.3	85.6	81.9	76.6	74.1	(28)
	2%	WB	64.5	66.3	66.8	72.1	74.9	79.7	80.5	80.3	78.6	76.3	71.1	69.9	(29)
		MCDB	68.0	71.4	73.0	77.4	81.6	87.6	87.9	86.6	83.6	81.3	72.7	71.0	(30)
	5%	WB	62.7	64.0	65.0	70.6	73.8	78.4	79.5	79.5	77.8	74.7	68.7	67.2	(31)
		MCDB	64.6	67.5	70.4	75.6	79.9	86.2	86.6	86.0	82.9	80.2	71.2	68.8	(32)
	10%	WB	60.1	61.9	63.3	69.0	72.9	77.5	78.6	78.6	76.8	73.1	66.0	63.6	(33)
		MCDB	63.0	65.5	67.4	73.9	78.9	85.2	85.7	84.7	82.1	77.7	70.4	65.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	17.8	17.7	16.6	15.0	14.5	14.0	13.3	13.4	15.8	16.9	16.0	(35)
		MCDBR	19.2	19.8	20.2	17.7	17.6	17.1	15.5	15.3	15.1	15.9	18.0	17.0	(36)
		MCWBR	12.7	12.4	10.7	7.9	6.6	6.8	6.6	6.1	5.4	7.7	10.9	11.8	(37)
	5% WB	MCDBR	16.2	17.2	17.2	14.1	13.9	14.9	14.2	14.0	12.9	13.7	13.7	14.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.339	0.369	0.402	0.454	0.493	0.522	0.505	0.429	0.394	0.353	0.338	(40)
		taud	2.534	2.505	2.441	2.340	2.234	2.173	2.107	2.156	2.364	2.431	2.502	2.539	(41)
		Ebn at Noon	283	291	289	282	267	255	247	250	266	268	272	273	(42)
	Edh at Noon		27	30	35	40	45	48	50	48	37	32	28	25	(43)
(44) All-Sky Solar Radiation	RadAvg		930	1140	1492	1895	2028	2034	1978	1792	1520	1327	1035	826	(44)
	RadStd		49	115	136	121	133	145	121	118	126	139	70	66	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (36 neighbors)	+1.08	+1.37	N/A	N/A	N/A	N/A	N/A	-176	+374	+259	(47)

Nomenclature: See separate page